

Nonquadratic Fixed Points of $f = \tau f^*$

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Source Problem

The source paper is:

A. N. Iusem, D. Reem, and S. Reich, *Fixed points of Legendre-Fenchel type transforms*, arXiv:1708.00464.

Remark 15.1 asks whether the fixed point equation

$$f(x) = \tau f^*(Ex + c) + \langle w, x \rangle + \beta$$

always has a unique solution when E is positive definite. The authors explicitly mention that even the simple case

$$f(x) = \tau f^*(x), \quad 0 < \tau \neq 1,$$

is unclear, and they suspect non-uniqueness.

15. CONCLUDING REMARKS AND OPEN PROBLEMS

We conclude the paper with the following remarks.

Remark 15.1. At the moment it is not clear whether (1.1) always has a unique solution (namely, the quadratic function with coefficients defined in (3.1)) whenever E is positive definite. Actually, even in the simple cases, where $0 < \tau \neq 1$ and $f(x) = \tau f^*(x)$ for all $x \in X$ or $f(x) = f^*(x + c)$ for each $x \in X$, where $c \neq 0$ is fixed, it is not clear whether the above-mentioned quadratic function is the unique solution to one of these equations, and we suspect that non-uniqueness can hold. As we saw in Sections 11–13, even more substantial complications arise when E is not positive definite, and the task of giving a complete description of the structure of the solutions to (1.1) in this case, as a function of the various parameters which appear in (1.1), seems to be out of reach now even in the finite-dimensional case

Result

Theorem 1. *For every $\tau > 0$, $\tau \neq 1$, there exists a nonquadratic, differentiable, proper, lower semicontinuous convex function*

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

such that

$$f = \tau f^*.$$

Thus the positive-definite uniqueness question in Remark 15.1 has a negative answer already in the one-dimensional equation $f(x) = \tau f^(x)$.*

Proof Intuition

If f is differentiable and strictly convex with derivative $h = f'$, then $(f^*)' = h^{-1}$. Differentiating $f = \tau f^*$ suggests the relation

$$h(x) = \tau h^{-1}(x).$$

Conversely, if an increasing homeomorphism h satisfies

$$h^{-1}(x) = h(x)/\tau,$$

then $f(x) = \int_0^x h(s) ds$ satisfies $f = \tau f^*$, after the harmless normalization $f(0) = 0$.

The construction of h is easiest in logarithmic coordinates on $(0, \infty)$. The equation becomes the problem of taking an increasing square root of a translation. Translation has many non-linear increasing square roots: conjugate the half-translation by a nontrivial periodic perturbation of the identity.

Construction of the Derivative

Let

$$a = \log \tau \neq 0.$$

Choose a real number $\varepsilon \neq 0$ so small that

$$|\varepsilon| < \frac{|a|}{2\pi}.$$

Define

$$H(u) = u + \varepsilon \sin\left(\frac{2\pi u}{a}\right), \quad u \in \mathbb{R}.$$

Then H is a strictly increasing homeomorphism of $\mathbb{R}$, and

$$H(u + a) = H(u) + a.$$

Set

$$R = H^{-1} \circ T_{-a/2} \circ H, \quad T_b(u) = u + b.$$

Since H commutes with T_a , we have

$$R^2 = H^{-1} T_{-a} H = T_{-a}.$$

Now define, for $x > 0$,

$$h(x) = \exp(R(\log x) + a),$$

set $h(0) = 0$, and extend h oddly:

$$h(-x) = -h(x) \quad (x > 0).$$

Lemma 1. *The function $h : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing homeomorphism and satisfies*

$$h^{-1}(x) = \frac{h(x)}{\tau} \quad (x \in \mathbb{R}).$$

Proof. On $(0, \infty)$, the function $x \mapsto \log x$, followed by R , followed by $u \mapsto \exp(u + a)$, is a strictly increasing homeomorphism from $(0, \infty)$ onto $(0, \infty)$. Since $R(u) = u - a/2 + O(1)$ as $u \rightarrow \pm\infty$, we have $h(x) \rightarrow 0$ as $x \downarrow 0$ and $h(x) \rightarrow \infty$ as $x \rightarrow \infty$. The odd extension is therefore a strictly increasing homeomorphism of $\mathbb{R}$.

Let $x = e^u > 0$. Then

$$\frac{h(x)}{\tau} = \exp(R(u) + a - a) = \exp(R(u)).$$

Using $R^2(u) = u - a$, we get

$$h\left(\frac{h(x)}{\tau}\right) = h(e^{R(u)}) = \exp(R(R(u)) + a) = \exp(u - a + a) = x.$$

Thus $h(h(x)/\tau) = x$ for $x > 0$. By oddness the same identity holds for all $x \in \mathbb{R}$. Since h is bijective, this is equivalent to

$$h^{-1}(x) = h(x)/\tau.$$

□

Proof of the Fixed Point Equation

Define

$$f(x) = \int_0^x h(s) ds, \quad x \in \mathbb{R}.$$

Since h is continuous and increasing, f is differentiable, convex, and lower semicontinuous. Since h is onto, f is superlinear at both infinities; in particular f^* is finite everywhere. Moreover $f(0) = 0$ and 0 is the unique minimizer, so $f^*(0) = 0$.

Because $f' = h$, the derivative of the convex conjugate is

$$(f^*)' = h^{-1}.$$

Therefore the function

$$\Phi(x) = f(x) - \tau f^*(x)$$

has derivative

$$\Phi'(x) = h(x) - \tau h^{-1}(x) = 0$$

for every x . Hence Φ is constant. At $x = 0$,

$$\Phi(0) = f(0) - \tau f^*(0) = 0,$$

so $\Phi \equiv 0$, which is exactly

$$f = \tau f^*.$$

Finally, f is not quadratic. If it were quadratic, then its derivative h would be linear. On $(0, \infty)$, linearity of h would force $R(u) + a - u$ to be constant, equivalently $R = T_{-a/2}$. But

$$R = H^{-1}T_{-a/2}H$$

equals $T_{-a/2}$ only if H commutes with $T_{-a/2}$. The chosen

$$H(u) = u + \varepsilon \sin(2\pi u/a)$$

does not commute with $T_{-a/2}$ when $\varepsilon \neq 0$, since the sine term changes sign under a half-period translation. Thus h is non-linear and f is nonquadratic.

Verification Notes

The proof gives a continuum of examples, one for each nontrivial periodic perturbation H . The construction is normalized by $f(0) = 0$; adding a constant would destroy the equation unless the constant is zero.

The result is a counterexample to uniqueness in the positive-definite case

$$E = \text{Id}, \quad c = w = 0, \quad \beta = 0, \quad 0 < \tau \neq 1.$$

It does not attempt to classify all solutions of the general equation with arbitrary E, c, w, β .

Novelty Check

On 2026-06-28, I searched for exact and close variants of the simple equation and the source-paper question:

`"f = tau f*" Legendre Fenchel nonunique solution,`

`"f(x)=\tau f*(x)" convex conjugate nonquadratic,`

`"Fixed points of Legendre-Fenchel type transforms" "non-uniqueness" "tau",`

and

`"f=alpha f*" "convex conjugate".`

No explicit later answer to this positive-definite uniqueness question was found in that bounded search.

References

References

- [1] A. N. Iusem, D. Reem, and S. Reich, *Fixed points of Legendre-Fenchel type transforms*, arXiv:1708.00464.